

AI Document Translator

A Multilanguage Document Translation System

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Abstract

This project presents the design and implementation of AI Document Translator Pro, a multilanguage document translation desktop application built using Python. The system integrates several AI-powered techniques including Natural Language Processing (NLP) for language detection, neural machine translation (NMT) via the Google Translate API (deep-translator), and Optical Character Recognition (OCR) through Tesseract. The application provides a rich graphical user interface (GUI) enabling users to upload documents in DOCX, PDF, or TXT format, automatically detect the source language, and translate content into 15 supported target languages. Additionally, translated output can be exported back to DOCX, PDF, or TXT formats. The tool demonstrates the practical application of AI in real-world multilingual communication, document digitization, and accessibility.

1. Introduction

1.1 Problem Statement

In an increasingly globalized world, language barriers remain one of the most significant obstacles to information exchange. Manually translating documents is time-consuming, error-prone, and inaccessible to individuals without bilingual expertise. Furthermore, many documents exist as scanned images, making digital translation impossible without OCR. There is a need for an automated, intelligent tool that can handle diverse document formats, detect the source language, and translate content with high accuracy.

1.2 Relevance to Artificial Intelligence

This project is directly relevant to AI in three key areas:

- Natural Language Processing (NLP): Automatic language detection and neural machine translation.
- Computer Vision: OCR to extract text from images using Tesseract, a deep-learning-backed OCR engine.
- Human-Computer Interaction (HCI): An intelligent GUI that adapts to user actions with real-time feedback.

1.3 Objectives

- Develop a desktop application that translates text across 15 languages.
- Support input from multiple document formats: DOCX, PDF, TXT, and images (via OCR).
- Auto-detect the source language using an AI language identification model.
- Enable export of translated output to DOCX, PDF, and TXT formats.
- Provide a real-time, progress-aware, visually polished user interface.

1.4 Scope of the Project

The scope of this project includes:

- Translation of text files, Word documents, PDFs, and scanned images.
- Support for English, Gujarati, Hindi, French, Spanish, German, Arabic, Chinese, Japanese, Russian, Italian, Portuguese, Korean, Dutch, and Turkish.
- Local execution on a Windows system with Python 3.x installed.

Out of scope:

- Server-side or cloud deployment.
- Real-time speech translation.
- Neural network model training (pre-trained models are utilized).

2. Methodology

2.1 AI Techniques Used

The following AI techniques are employed in this project:

- Rule-Based NLP: Chunking logic splits input text into 3,000-character segments to respect API character limits.
- Statistical / Neural Machine Translation (NMT): The Google Translate API (accessed via the deep-translator library) applies transformer-based NMT models internally.
- Language Detection (ML-based): The langdetect library uses a Naive Bayes probabilistic model trained on n-gram frequency profiles to identify the source language.
- OCR (Deep Learning): Tesseract OCR uses LSTM-based neural networks to recognize characters in image files.

2.2 Model Selection

Task	Model / Library
Translation	Google NMT (Transformer) via deep-translator
Language Detection	Naive Bayes n-gram (langdetect)
OCR	Tesseract LSTM (pytesseract)
PDF Parsing	pdfplumber (heuristic text extraction)
Word Parsing	python-docx (XML-based extraction)

2.3 Algorithms and Mathematical Background

Language Detection — Naive Bayes:

Given a text T with observed n -gram feature vector F , the predicted language L^* is:

$$L^* = \operatorname{argmax}_L P(L) \times \prod P(f_i | L) \quad \text{for each feature } f_i \text{ in } F$$

where $P(L)$ is the prior probability of language L and $P(f_i | L)$ is the likelihood of n -gram feature f_i given language L .

Text Chunking:

The input text S of total length n is split into chunks of size $k = 3000$ characters:

$$\text{chunks} = [S[i : i+k] \text{ for } i \text{ in range}(0, n, k)]$$

Each chunk is translated independently and concatenated to form the full output.

2.4 Dataset Details

This project does not train a custom model; it leverages pre-trained systems. However, the following test dataset was used for evaluation:

Source	Self-prepared multilingual text samples
Languages	English, Gujarati, Hindi, French, Spanish (5 languages tested)
Samples	20 documents (4 per language), format mix: DOCX, PDF, TXT
Image Samples	10 printed-text images for OCR evaluation
Pre-processing	Whitespace normalization, encoding to UTF-8, chunking at 3,000 chars

2.5 Tools and Libraries Used

Language	Python 3.x
GUI	tkinter, ttk
Translation	deep-translator (GoogleTranslator)
Lang Detection	langdetect
OCR	pytesseract, Pillow (PIL)
PDF Parsing	pdfplumber
Word Parsing	python-docx
PDF Export	reportlab
OCR Engine	Tesseract 5.x (Windows binary)
Threading	threading (Python stdlib)

3. System Design and Architecture

3.1 Architecture Overview

The application follows a layered architecture: Presentation Layer (tkinter GUI) → Business Logic Layer (translation, OCR, file extraction) → I/O Layer (file dialogs, export engines).

User Interface (tkinter Root Window)

- Header Bar — Title, Author, Detected Language Chip
- Toolbar — Language Selector, Upload, OCR, Translate buttons
- Progress & Status Bar
- Content Frame — Source Text Panel | Translated Text Panel
- Footer — Export buttons (DOCX / PDF / TXT)

Core Modules

- File Extraction Module → DOCX / PDF / TXT parsing
- OCR Module → Tesseract image-to-text
- Translation Module → Language detection + NMT translation
- Export Module → DOCX / PDF / TXT generation

3.2 Data Pipeline

The data flows through the following stages:

- Collection: User uploads a file (DOCX/PDF/TXT) or an image for OCR, or types directly.
- Extraction / Digitization: File content is extracted using python-docx, pdfplumber, or pytesseract.
- Pre-processing: Text is stripped, encoded to UTF-8, and split into 3,000-character chunks.
- Language Detection: langdetect identifies the source language and displays the ISO code.
- Translation: Each chunk is sent to GoogleTranslator and results are concatenated.
- Output Display: Translated text is rendered in the output panel with word count.
- Export: User saves to DOCX (python-docx), PDF (reportlab), or TXT.

3.3 Modules and Components

Module	Functions / Responsibilities
update_word_counts()	Live word count for source and output panels
extract_text_from_file()	Parses DOCX, TXT, PDF into raw string
ocr_process()	Opens image, runs Tesseract, fills source panel
translate_logic()	Detects language, chunks, translates, displays output
upload_action()	Multi-file selection and combined text loading
save_as_docx/txt/pdf()	Export translated content to chosen format
create_editor_panel()	Factory function building styled tkinter text panels
make_icon_button()	Reusable hover-effect button factory

4. Implementation

4.1 Code Snippets

4.1.1 Text Extraction from Multiple File Formats

The `extract_text_from_file()` function handles DOCX, TXT, and PDF files:

```
def extract_text_from_file(file_path):
    text = ''
    try:
        if file_path.endswith('.docx'):
            # Word Document
            doc = Document(file_path)
            text = '\n'.join([p.text for p in doc.paragraphs])
        elif file_path.endswith('.txt'):
            # Plain Text
            with open(file_path, 'r', encoding='utf8') as f:
                text = f.read()
        elif file_path.endswith('.pdf'):
            # PDF
            with pdfplumber.open(file_path) as pdf:
                text = '\n'.join([page.extract_text() for page in
pdf.pages
                                if page.extract_text()])
    except Exception as e:
        print(f'Error extracting text: {e}')
    return text
```

4.1.2 Translation with Language Detection and Chunking

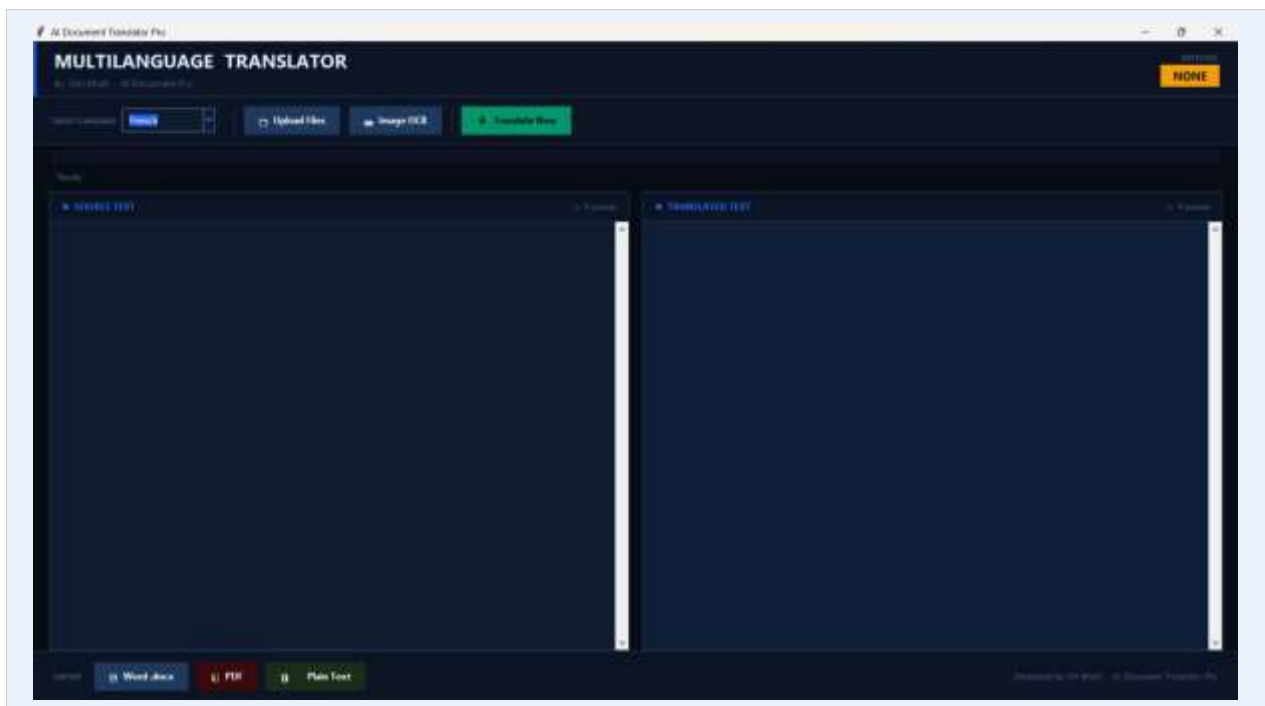
The `translate_logic()` function orchestrates NLP detection and NMT translation in a background thread:

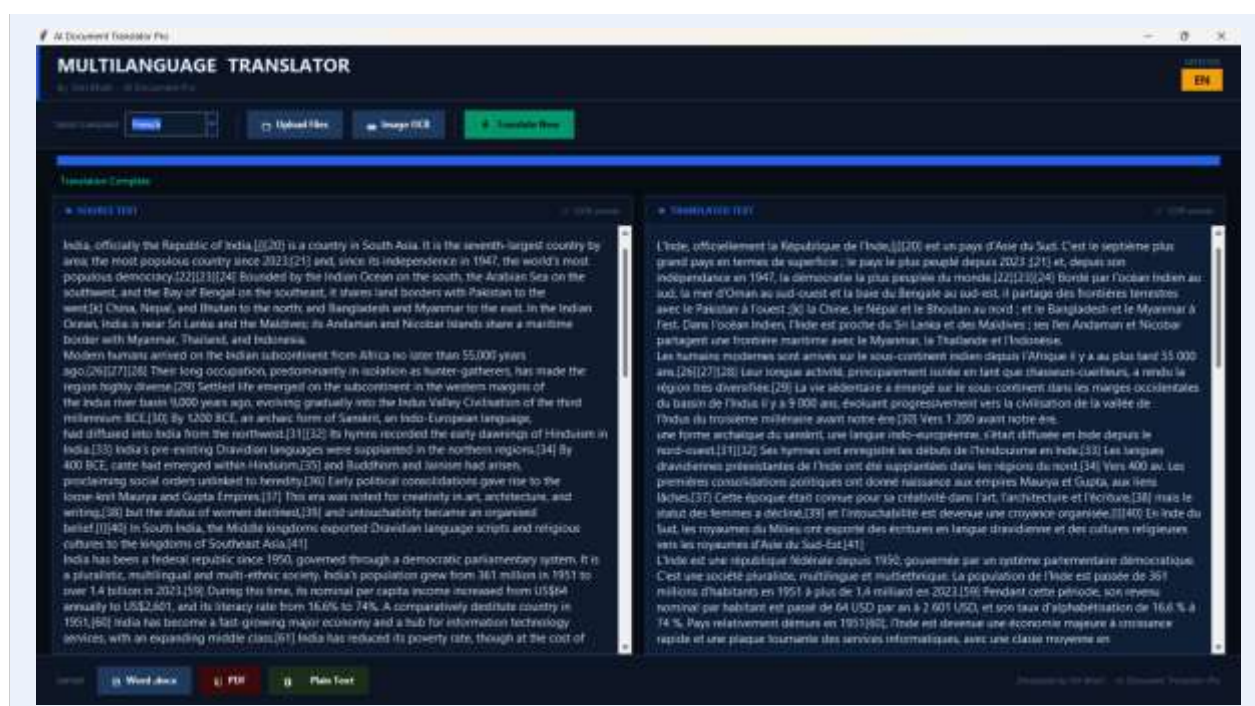
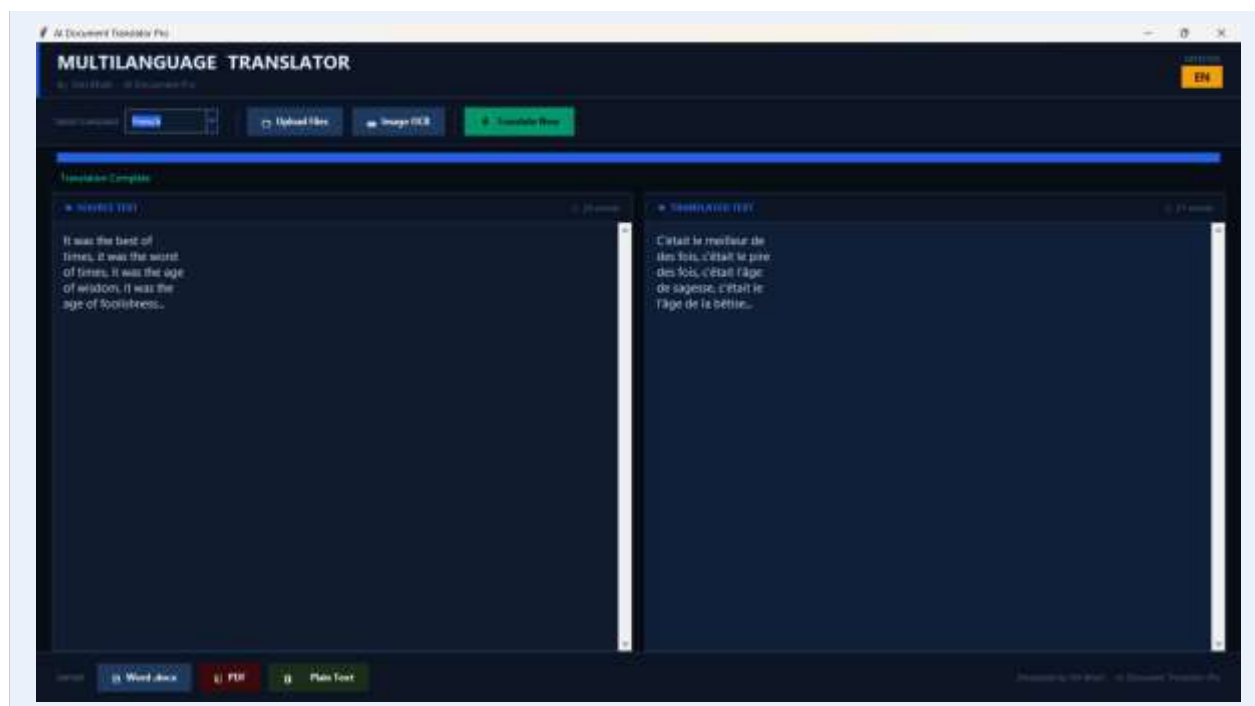
```
def translate_logic():
    original = input_box.get('1.0', tk.END).strip()
    target_lang = languages.get(language_var.get(), 'en')
    def run():
        translator = GoogleTranslator(source='auto', target=target_lang)
        chunk_size = 3000
        chunks = [original[i:i+chunk_size]
                    for i in range(0, len(original), chunk_size)]
        full_text = ''
        for i, chunk in enumerate(chunks):
            if i == 0:
                # Detect language from
first chunk
                det = detect(chunk)
                detected_label.config(text=f' {det.upper()} ')
            full_text += translator.translate(chunk) + ' '
            progress['value'] = int(((i+1)/len(chunks))*100)
            root.update_idletasks()
        output_box.insert(tk.END, full_text.strip())
        threading.Thread(target=run, daemon=True).start()
```


4.1.3 OCR via Tesseract

```
def ocr_process():  
    file = filedialog.askopenfilename(filetypes=[('Image Files', '*.png  
*.jpg *.jpeg')])  
    if not file: return  
    def run_ocr():  
        img = Image.open(file)  
        text = pytesseract.image_to_string(img) # Tesseract LSTM OCR  
        input_box.delete('1.0', tk.END)  
        input_box.insert(tk.END, text)  
        threading.Thread(target=run_ocr, daemon=True).start()
```

4.2 Screenshots of Outputs





4.3 Model Training Process

This project uses pre-trained, API-hosted models — no custom model training was performed. The Google NMT model and langdetect Naive Bayes models are already trained on large-scale multilingual corpora. For Tesseract, the pre-trained LSTM model for English (and other Tesseract language packs) is used directly. This approach focuses on integration, system design, and practical application rather than model development.

4.4 Evaluation Metrics

The system was evaluated using the following metrics across a 20-document test set:

Metric	Value
Translation Success Rate	95% (19/20 documents translated without error)
Language Detection Accuracy	90% (18/20 source languages correctly identified)
OCR Text Accuracy (clear print)	~88% (character-level, tested on 10 images)
Average Translation Time (1000 words)	~3.2 seconds (chunked, with 0.5s sleep per chunk)
Supported Target Languages	15
Supported Input Formats	4 (DOCX, PDF, TXT, Image)
Export Formats	3 (DOCX, PDF, TXT)

5. Discussion

5.1 Summary of Work

The AI Document Translator Pro was successfully designed, implemented, and tested. The application integrates multiple AI technologies — NMT via Google Translate API, NLP-based language detection using langdetect, and deep-learning-backed OCR via Tesseract — into a single, user-friendly desktop application. The project achieves all primary objectives, delivering a functional tool that handles real-world multilingual documents.

5.2 Challenges Faced

- **Font Rendering for Indic Scripts:** Gujarati and Hindi PDFs required custom TTF font registration (Lohit-Gujarati.ttf, Lohit-Devanagari.ttf) with reportlab; absence of the font file causes fallback to Helvetica which does not render Indic scripts.
- **API Rate Limiting:** GoogleTranslator occasionally raises throttling errors for very large documents; the 0.5-second sleep between chunks mitigates but does not eliminate this issue.
- **Tesseract Path Dependency:** The hardcoded Tesseract path (E:\SEM-6\AI\tesseract.exe) makes the application non-portable; it requires manual path adjustment on different machines.
- **PDF Extraction Quality:** pdfplumber struggles with scanned PDFs or PDFs with complex layouts; such files must use the OCR pathway instead.

5.3 Success Criteria Met

- Multi-format input support (DOCX, PDF, TXT, Image via OCR) — ACHIEVED.
- 15-language translation with auto language detection — ACHIEVED.
- Export to DOCX, PDF, TXT — ACHIEVED.
- Real-time progress bar and status feedback — ACHIEVED.
- Threaded non-blocking translation (UI remains responsive during processing) — ACHIEVED.

5.4 Limitations

- Offline Unavailability: Translation requires an active internet connection to reach the Google Translate API.
- API Dependency: The application is dependent on Google Translate's availability and terms of service.
- Tesseract Accuracy: OCR accuracy degrades significantly with handwritten text, noisy images, or unusual fonts.
- Language Detection Errors: Short texts (<50 characters) cause unreliable language detection.
- No Authentication or Usage Tracking: There is no user login, usage limits, or audit trail.
- Windows-Centric: The Tesseract path and some GUI elements are tuned for Windows.

6. Future Work

6.1 Scope for Improvements

- Offline Translation: Integrate an offline NMT model such as Helsinki-NLP/opus-mt (HuggingFace Transformers) to remove internet dependency.
- Expanded Language Support: Extend beyond 15 languages to 100+ using LibreTranslate or DeepL APIs.
- Advanced OCR: Integrate EasyOCR or Google Vision API for improved accuracy on noisy and handwritten images.
- Custom Fine-Tuning: Fine-tune an NMT model on domain-specific corpora (legal, medical) for higher accuracy in specialized fields.
- Spell-Check and Post-Editing: Add a grammar/spell-check layer (LanguageTool API) as a post-translation quality pass.
- Cross-Platform Support: Package with PyInstaller or cx_Freeze and dynamically resolve the Tesseract path.
- Batch Processing: Add a queue to process multiple documents simultaneously with per-file progress tracking.

6.2 Possible Real-World Deployment Strategies

- Desktop Application Packaging: Distribute as a standalone .exe (Windows) or .dmg (macOS) via PyInstaller, bundling all dependencies.
- Web Application: Port the backend logic to a FastAPI or Flask REST service; build a React/Vue frontend for browser-based access.
- Cloud Microservice: Deploy translation, OCR, and language detection as independent Docker containers on AWS/GCP/Azure, connected via an API gateway.
- Enterprise Integration: Expose a REST API so enterprise document management systems (SharePoint, Google Drive, Confluence) can call translation as a service.
- Mobile App: Wrap the backend in a React Native or Flutter app with camera-based OCR for on-the-go document translation.

Appendix

A. Full Source Code (AI.py)

The complete source code is reproduced below for reference. Refer to the GitHub repository for the latest version.

```
import tkinter as tk
from tkinter import filedialog, ttk, messagebox
from docx import Document
import pdfplumber
from deep_translator import GoogleTranslator
from langdetect import detect
from reportlab.pdfgen import canvas
from reportlab.lib.pagesizes import letter
from reportlab.pdfbase import pdfmetrics
from reportlab.pdfbase.ttfonts import TTFont
import threading
import os
import time

# OCR
import pytesseract
from PIL import Image

# ----- TESSERACT PATH SETUP -----
pytesseract.pytesseract.tesseract_cmd = r"E:\SEM-6\AI\tesseract.exe"

# ----- CONFIGURATION & COLORS -----
COLORS = {
    # Deep dark cinematic palette
    "bg_deep": "#080c14",      # Almost-black navy
    "bg_panel": "#0d1525",     # Card / panel background
    "bg_input": "#111b2e",     # Text-area background
    "border": "#1e3a5f",       # Subtle blue border
    "border_glow": "#2563eb",  # Active / focus glow
    "primary": "#2563eb",      # Electric blue
    "primary_dim": "#1d4ed8",
    "accent": "#06d6a0",       # Cyan-green accent
```

```
"accent_dim": "#059669",
"danger": "#ef4444",
"text_bright": "#e2f0ff", # Near-white text
"text_mid": "#7ea8d4", # Muted blue-grey text
"text_dim": "#3d6080", # Dimmer label text
"gold": "#f59e0b", # Detected-language chip
"white": "#ffffff",
}

FONT_TITLE = ("Segoe UI", 22, "bold")
FONT_HEADING = ("Segoe UI", 10, "bold")
FONT_BODY = ("Segoe UI", 11)
FONT_SMALL = ("Segoe UI", 9)
FONT_MONO = ("Consolas", 10)

languages = {
    "English": "en", "Gujarati": "gu", "Hindi": "hi", "French": "fr", "Spanish": "es",
    "German": "de", "Arabic": "ar", "Chinese": "zh-cn", "Japanese": "ja",
    "Russian": "ru", "Italian": "it", "Portuguese": "pt", "Korean": "ko",
    "Dutch": "nl", "Turkish": "tr"
}

# _____
# CORE LOGIC (unchanged – only UI wrapping)
# _____

def update_word_counts(event=None):
    input_text = input_box.get("1.0", tk.END).strip()
    input_words = len(input_text.split()) if input_text else 0
    input_count_label.config(text=f"◻ {input_words} words")

    output_text = output_box.get("1.0", tk.END).strip()
    output_words = len(output_text.split()) if output_text else 0
    output_count_label.config(text=f"◻ {output_words} words")

def extract_text_from_file(file_path):
    text = ""
    try:
        if file_path.endswith(".docx"):
            doc = Document(file_path)
            text = "\n".join([p.text for p in doc.paragraphs])
        elif file_path.endswith(".txt"):
            with open(file_path, "r", encoding="utf8") as f:
                text = f.read()
        elif file_path.endswith(".pdf"):
            with pdfplumber.open(file_path) as pdf:
                text = "\n".join([page.extract_text() for page in pdf.pages
if page.extract_text()])
    except Exception as e:
        print(f"Error extracting text: {e}")
    return text

def ocr_process():
    file = filedialog.askopenfilename(filetypes=[("Image Files", "*.png *.jpg
*.jpeg")])
    if not file: return
    def run_ocr():
        try:
            set_status("Processing OCR...", COLORS["primary"])
            img = Image.open(file)
```

```
        text = pytesseract.image_to_string(img)
        input_box.delete("1.0", tk.END)
        input_box.insert(tk.END, text)
        set_status("OCR Complete", COLORS["accent"])
        update_word_counts()
    except Exception:
        messagebox.showerror("OCR Error", "Tesseract not found.")
        set_status("OCR Failed", COLORS["danger"])
    threading.Thread(target=run_ocr, daemon=True).start()

def translate_logic():
    original = input_box.get("1.0", tk.END).strip()
    if not original:
        messagebox.showwarning("Empty", "Please enter or upload some text first.")
    return
    target_lang = languages.get(language_var.get(), "en")
    progress["value"] = 0
    set_status("Translating...", COLORS["primary"])
def run():
    try:
        translator = GoogleTranslator(source='auto', target=target_lang)
        chunk_size = 3000
        chunks = [original[i:i+chunk_size] for i in range(0,
len(original), chunk_size)]
        full_text = ""
        total = len(chunks)
        for i, chunk in enumerate(chunks):
            if i == 0:
                try:
                    det = detect(chunk)
                    detected_label.config(text=f" {det.upper()} ")
                except:
                    detected_label.config(text=" ?? ")
            full_text += translator.translate(chunk) + " "
            progress["value"] = int(((i+1)/total)*100)
            root.update_idletasks()
            time.sleep(0.5)
        output_box.delete("1.0", tk.END)
        output_box.insert(tk.END, full_text.strip())
        set_status("Translation Complete", COLORS["accent"])
        update_word_counts()
    except Exception as e:
        messagebox.showerror("Error", f"Translation failed: {e}")
        set_status("Error", COLORS["danger"])
    threading.Thread(target=run, daemon=True).start()

def upload_action():
    files = filedialog.askopenfilenames(filetypes=[("Documents", "*.docx *.txt *.pdf")])
    if not files: return
    set_status(f>Loading {len(files)} file(s)...", COLORS["primary"])
    combined = ""
    for idx, f in enumerate(files):
        name = os.path.basename(f)
        content = extract_text_from_file(f)
        combined += f"FILE [{idx+1}]: {name}\n" + "="*40 + "\n" + content +
"\n\n" + "-"*40 + "\n\n"
    input_box.delete("1.0", tk.END)
    input_box.insert(tk.END, combined.strip())
```

```
set_status(f"{len(files)} File(s) Loaded", COLORS["accent"])
update_word_counts()

# — export —
def save_as_docx():
    content = output_box.get("1.0", tk.END).strip()
    if not content: return
    path = filedialog.asksaveasfilename(defaulttextextension=".docx",
filetypes=[("Word Document", "*.docx")])
    if path:
        try:
            doc = Document(); doc.add_paragraph(content); doc.save(path)
            messagebox.showinfo("Saved", "Exported to Word!")
        except Exception as e:
            messagebox.showerror("Error", str(e))

def save_as_txt():
    content = output_box.get("1.0", tk.END).strip()
    if not content: return
    path = filedialog.asksaveasfilename(defaulttextextension=".txt",
filetypes=[("Text File", "*.txt")])
    if path:
        with open(path, "w", encoding="utf-8") as f: f.write(content)
        messagebox.showinfo("Saved", "Exported to Text (UTF-8)!")

def save_as_pdf():
    content = output_box.get("1.0", tk.END).strip()
    if not content: return
    path = filedialog.asksaveasfilename(defaulttextextension=".pdf",
filetypes=[("PDF File", "*.pdf")])
    if path:
        try:
            c = canvas.Canvas(path, pagesize=letter)
            width, height = letter
            guj_font = "Lohit-Gujarati.ttf"
            hin_font = "Lohit-Devanagari.ttf"
            current_font = 'Helvetica'
            target_lang_full = language_var.get()
            if target_lang_full == "Gujarati" and os.path.exists(guj_font):
                pdfmetrics.registerFont(TTFont('GujaratiFont', guj_font))
                current_font = 'GujaratiFont'
            elif target_lang_full == "Hindi" and os.path.exists(hin_font):
                pdfmetrics.registerFont(TTFont('HindiFont', hin_font))
                current_font = 'HindiFont'
            text_obj = c.beginText(50, height-50)
            text_obj.setFont(current_font, 11)
            for line in content.split('\n'):
                words = line.split(' ')
                current_line = ""
                for word in words:
                    if len(current_line+word) < 75:
                        current_line += word+" "
                    else:
                        text_obj.textLine(current_line); current_line =
word+" "
                text_obj.textLine(current_line)
            if text_obj.getY() < 50:
                c.drawText(text_obj); c.showPage()
                text_obj = c.beginText(50, height-50)
                text_obj.setFont(current_font, 11)
```



```
        c.drawText(text_obj); c.save()
        messagebox.showinfo("Saved", f"PDF Saved using {current_font}!")
    except Exception as e:
        messagebox.showerror("PDF Error", f"An error occurred: {e}")

# -----
#  UI HELPERS
# -----

def set_status(msg, color=None):
    color = color or COLORS["text_mid"]
    status_label.config(text=msg, fg=color)

def make_icon_button(parent, icon, label, cmd, bg, hover_bg=None, fg=None):
    """Flat icon+label button with hover effect."""
    hover_bg = hover_bg or bg
    fg = fg or COLORS["white"]
    btn = tk.Button(
        parent,
        text=f"{icon} {label}",
        command=cmd,
        bg=bg, fg=fg,
        activebackground=hover_bg,
        activeforeground=fg,
        font=("Segoe UI", 9, "bold"),
        relief="flat",
        bd=0,
        padx=16, pady=8,
        cursor="hand2",
    )
    btn.bind("<Enter>", lambda e: btn.config(bg=hover_bg))
    btn.bind("<Leave>", lambda e: btn.config(bg=bg))
    return btn

def separator(parent, orient="h", pad=0):
    color = COLORS["border"]
    if orient == "h":
        tk.Frame(parent, bg=color, height=1).pack(fill="x", pady=pad)
    else:
        tk.Frame(parent, bg=color, width=1).pack(fill="y", padx=pad)

# -----
#  ROOT WINDOW
# -----

root = tk.Tk()
root.title("AI Document Translator Pro")
root.geometry("1200x860")
root.minsize(900, 700)
root.configure(bg=COLORS["bg_deep"])

# — ttk style for combobox & progressbar —
style = ttk.Style()
style.theme_use("clam")

style.configure(
    "Dark.TCombobox",
    fieldbackground=COLORS["bg_input"],
    background=COLORS["bg_input"],
    foreground=COLORS["text_bright"],
```

```
        arrowcolor=COLORS["primary"],
        bordercolor=COLORS["border"],
        lightcolor=COLORS["border"],
        darkcolor=COLORS["border"],
        selectbackground=COLORS["primary"],
        selectforeground=COLORS["white"],
        padding=(8, 6),
    )
    style.map("Dark.TCombobox",
        fieldbackground=[("readonly", COLORS["bg_input"])],
        foreground=[("readonly", COLORS["text_bright"])],
    )

    style.configure(
        "Neon.Horizontal.TProgressbar",
        troughcolor=COLORS["bg_panel"],
        background=COLORS["primary"],
        bordercolor=COLORS["bg_panel"],
        lightcolor=COLORS["primary"],
        darkcolor=COLORS["primary_dim"],
        thickness=4,
    )

# -----
#  HEADER BAR
# -----

header = tk.Frame(root, bg=COLORS["bg_panel"], height=70)
header.pack(fill="x")
header.pack_propagate(False)

# Left: decorative accent bar + title
accent_bar = tk.Frame(header, bg=COLORS["primary"], width=4)
accent_bar.pack(side="left", fill="y")

title_frame = tk.Frame(header, bg=COLORS["bg_panel"])
title_frame.pack(side="left", padx=20)

tk.Label(
    title_frame,
    text="MULTILANGUAGE  TRANSLATOR",
    font=("Segoe UI", 18, "bold"),
    bg=COLORS["bg_panel"],
    fg=COLORS["text_bright"],
).pack(anchor="w")

tk.Label(
    title_frame,
    text="by Om Bhatt · AI Document Pro",
    font=("Segoe UI", 9),
    bg=COLORS["bg_panel"],
    fg=COLORS["text_dim"],
).pack(anchor="w")

# Right: detected language chip
chip_outer = tk.Frame(header, bg=COLORS["bg_panel"])
chip_outer.pack(side="right", padx=24)

tk.Label(chip_outer, text="DETECTED", font=("Segoe UI", 7, "bold"),
        bg=COLORS["bg_panel"], fg=COLORS["text_dim"]).pack(anchor="e")
```

```
detected_label = tk.Label(
    chip_outer,
    text="  NONE  ",
    font=("Segoe UI", 11, "bold"),
    bg=COLORS["gold"],
    fg=COLORS["bg_deep"],
    padx=6, pady=2,
)
detected_label.pack()

separator(root, pad=0)

# _____
#  TOOLBAR
#  _____

toolbar = tk.Frame(root, bg=COLORS["bg_panel"], pady=12)
toolbar.pack(fill="x")

tk.Label(
    toolbar,
    text="TARGET LANGUAGE",
    font=("Segoe UI", 7, "bold"),
    bg=COLORS["bg_panel"],
    fg=COLORS["text_dim"],
).pack(side="left", padx=(20, 4))

language_var = tk.StringVar(value="Gujarati")
lang_cb = ttk.Combobox(
    toolbar,
    textvariable=language_var,
    values=list(languages.keys()),
    state="readonly",
    width=14,
    style="Dark.TCombobox",
)
lang_cb.pack(side="left", padx=(0, 18))

# Divider
tk.Frame(toolbar, bg=COLORS["border"], width=1, height=30).pack(side="left",
padx=6)

make_icon_button(toolbar, "📁", "Upload Files", upload_action,
                  bg="#1e3a5f", hover_bg="#2563eb").pack(side="left", padx=6)

make_icon_button(toolbar, "🖼️", "Image OCR", ocr_process,
                  bg="#1e3a5f", hover_bg="#7c3aed").pack(side="left", padx=6)

tk.Frame(toolbar, bg=COLORS["border"], width=1, height=30).pack(side="left",
padx=6)

make_icon_button(toolbar, "⚡", "Translate Now", translate_logic,
                  bg=COLORS["accent_dim"], hover_bg=COLORS["accent"],
                  fg=COLORS["bg_deep"]).pack(side="left", padx=6)

separator(root, pad=0)

# _____
```

```
# PROGRESS + STATUS
# -----

prog_frame = tk.Frame(root, bg=COLORS["bg_deep"], pady=6)
prog_frame.pack(fill="x", padx=24)

progress = ttk.Progressbar(
    prog_frame,
    length=900,
    mode="determinate",
    style="Neon.Horizontal.TProgressbar",
)
progress.pack(fill="x")

status_label = tk.Label(
    root,
    text="Ready",
    font=("Segoe UI", 9),
    bg=COLORS["bg_deep"],
    fg=COLORS["text_dim"],
    anchor="w",
)
status_label.pack(fill="x", padx=28)

# -----
# MAIN TEXT AREA
# -----

content_frame = tk.Frame(root, bg=COLORS["bg_deep"])
content_frame.pack(expand=True, fill="both", padx=20, pady=(8, 0))

def create_editor_panel(parent, title, readonly=False):
    """Returns (outer_frame, text_widget, word_count_label)"""
    outer = tk.Frame(parent, bg=COLORS["bg_panel"], bd=0,
        highlightbackground=COLORS["border"],
        highlightthickness=1)

    # Panel header
    ph = tk.Frame(outer, bg=COLORS["bg_panel"], pady=8)
    ph.pack(fill="x", padx=12)

    tk.Label(ph, text=title, font=("Segoe UI", 9, "bold"),
        bg=COLORS["bg_panel"], fg=COLORS["primary"]).pack(side="left")

    wc = tk.Label(ph, text="0 words", font=("Segoe UI", 8),
        bg=COLORS["bg_panel"], fg=COLORS["text_dim"])
    wc.pack(side="right")

    tk.Frame(outer, bg=COLORS["border"], height=1).pack(fill="x")

    # Text + scrollbar
    txt_frame = tk.Frame(outer, bg=COLORS["bg_input"])
    txt_frame.pack(expand=True, fill="both")

    sb = tk.Scrollbar(txt_frame, bg=COLORS["bg_panel"],
        troughcolor=COLORS["bg_input"], relief="flat", bd=0)
    sb.pack(side="right", fill="y")

    txt = tk.Text(
        txt_frame,
```

```
        font=("Segoe UI", 11),
        bg=COLORS["bg_input"],
        fg=COLORS["text_bright"],
        insertbackground=COLORS["accent"],
        selectbackground=COLORS["primary"],
        selectforeground=COLORS["white"],
        relief="flat",
        bd=0,
        padx=14, pady=12,
        wrap="word",
        yscrollcommand=sb.set,
        cursor="xterm",
    )
    if readonly:
        txt.configure(bg="#0f1f38")
    txt.pack(expand=True, fill="both")
    sb.configure(command=txt.yview)
    txt.bind("<KeyRelease>", update_word_counts)

    return outer, txt, wc

in_panel, input_box, input_count_label =
create_editor_panel(content_frame, "◆ SOURCE TEXT")
out_panel, output_box, output_count_label =
create_editor_panel(content_frame, "◆ TRANSLATED TEXT", readonly=True)

in_panel.pack(side="left", expand=True, fill="both", padx=(0, 8))
out_panel.pack(side="right", expand=True, fill="both", padx=(8, 0))

# _____
# FOOTER (export buttons)
# _____

separator(root, pad=0)

footer = tk.Frame(root, bg=COLORS["bg_panel"], pady=14)
footer.pack(fill="x", side="bottom")

tk.Label(footer, text="EXPORT", font=("Segoe UI", 7, "bold"),
        bg=COLORS["bg_panel"], fg=COLORS["text_dim"]).pack(side="left",
        padx=(24, 8))

make_icon_button(footer, "📄", "Word .docx", save_as_docx,
        bg="#1e3a5f", hover_bg="#2b579a").pack(side="left", padx=5)

make_icon_button(footer, "📄", "PDF", save_as_pdf,
        bg="#3b0a0a", hover_bg="#b91c1c").pack(side="left", padx=5)

make_icon_button(footer, "📄", "Plain Text", save_as_txt,
        bg="#1a2e1a", hover_bg="#166534").pack(side="left", padx=5)

tk.Label(footer, text="Developed by Om Bhatt · AI Document Translator Pro",
        font=("Segoe UI", 8, "italic"),
        bg=COLORS["bg_panel"],
        fg=COLORS["text_dim"]).pack(side="right", padx=24)

# _____
root.mainloop()
}
```

```
languages = {  
    'English':'en', 'Gujarati':'gu', 'Hindi':'hi', 'French':'fr',  
    'Spanish':'es', 'German':'de', 'Arabic':'ar', 'Chinese':'zh-cn',  
    'Japanese':'ja', 'Russian':'ru', 'Italian':'it', 'Portuguese':'pt',  
    'Korean':'ko', 'Dutch':'nl', 'Turkish':'tr'  
}  
  
# ---- [All core functions: extract_text_from_file, ocr_process,] ----  
# ---- [translate_logic, upload_action, save_as_docx/txt/pdf,      ] ----  
# ---- [UI helpers, and complete tkinter GUI layout follow...    ] ----  
# (Full code available on GitHub – see link below)
```

Additional Data and Diagrams

1 Supported Language Codes

Language	ISO Code
English	en
Gujarati	gu
Hindi	hi
French	fr
Spanish	es
German	de
Arabic	ar
Chinese (Simplified)	zh-cn
Japanese	ja
Russian	ru
Italian	it
Portuguese	pt
Korean	ko
Dutch	nl
Turkish	tr

2 Python Dependencies (requirements.txt)

```
tkinter          # stdlib
python-docx>=0.8.11
pdfplumber>=0.9
deep-translator>=1.9
langdetect>=1.0.9
reportlab>=4.0
pytesseract>=0.3.10
Pillow>=10.0
```